

human being.

To build the project, we require the components mentioned under the Bill of Material table. We will

also need some other parts for making this robot. Check November 2020 issue for that or the link <https://www.electronicsforu.com/>

*electronics-projects/design-iot-face-recognition-ai-robot*

This project has two distinct features, for which we need to create a basic setup for both face recognition and deep dreams.

For face recognition requirements/setup, we need to set up the following libraries in Raspberry Pi:

- Keras
- OpenCV
- eSpeak
- Dlib
- Face Recognition

To install the above libraries, open the terminal window in Raspberry Pi and run:

```
sudo apt-get update
sudo apt-get upgrade
sudo apt-get install build-essential
cmake pkg-config
sudo apt-get install libjpeg-dev
libtiff5-dev libjasper-dev libpng12-dev
sudo apt-get install libavcodec-dev
libavformat-dev libswscale-dev libv4l-dev
sudo apt-get install libxvidcore-dev
libx264-dev
sudo apt-get install libgtk2.0-dev
```

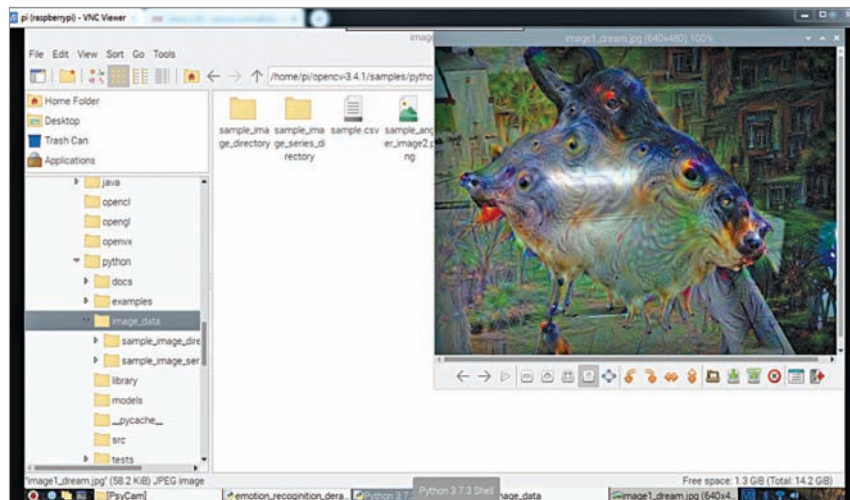


Fig. 3: Creating a dream

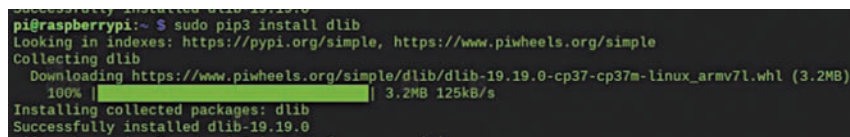


Fig. 4: Installing dlib

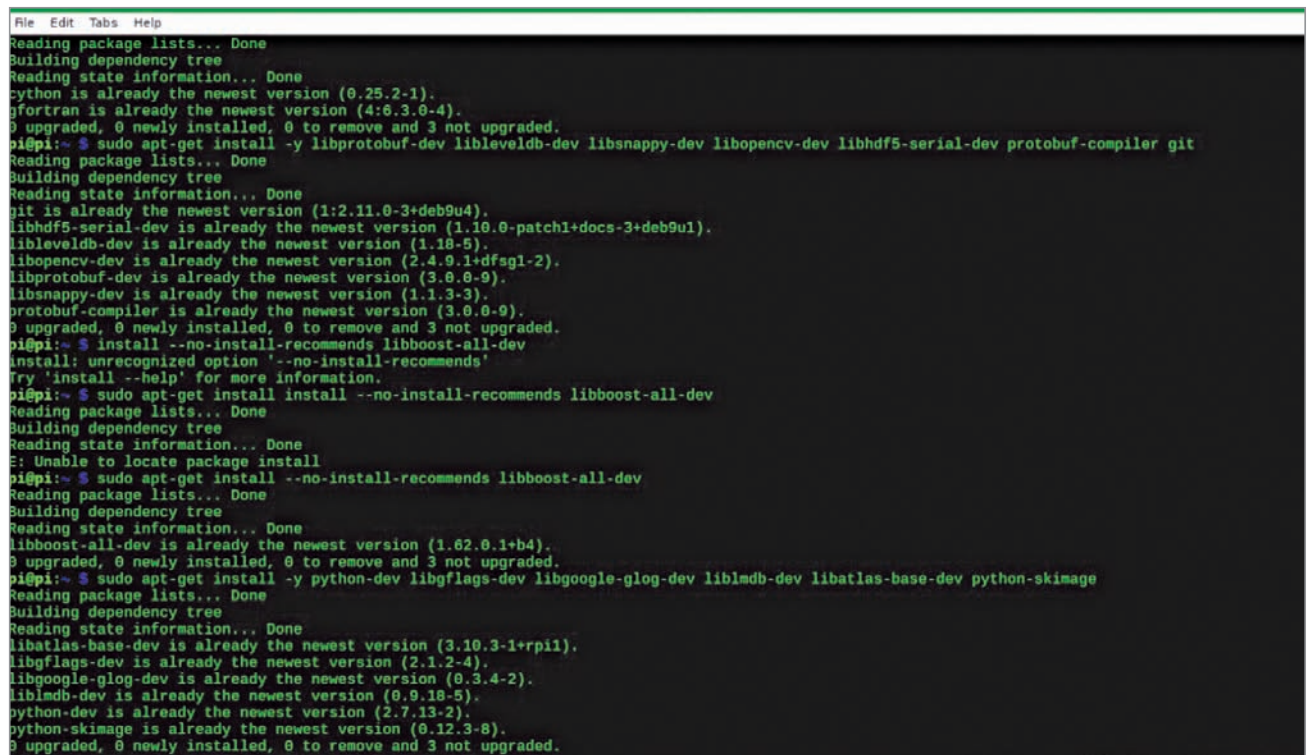


Fig. 5: Installing Python modules